DSA REFERENCE

# Breadth-First Search Problem Ladder

A sequenced practice ladder from warm-up to interview stretch.

Explore a graph in distance layers using a queue, which makes the first visit shortest in an unweighted graph.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

DSA / BFS

DSA

# Breadth-First Search Problem Ladder

**CORE IDEA**

**Explore a graph in distance layers using a queue, which makes the first visit shortest in an unweighted graph.**

### Recognition cues

+ minimum number of edges
+ level-order traversal
+ multi-source distance
+ nearest target

### Implementation sequence

+ mark on enqueue
+ process by layer when depth matters
+ seed all sources together
+ store parent only if reconstruction is needed

## Invariant

The queue contains discovered nodes whose shortest distance is known; later layers cannot improve earlier ones.

## Worked reasoning

For rotten oranges or nearest gate, enqueue all sources at distance zero and expand simultaneously.

## Complexity

O(V + E) time and O(V) queue/visited space.

## Failure modes

- [ ] marking on dequeue and duplicating work
- [ ] using BFS on weighted edges
- [ ] losing layer boundaries
- [ ] starting separate BFS from every source

Practice ladder

# Breadth-First Search Problem Ladder: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

| # | Problem | Deliberate-practice prompt |
|---|---|---|
| 1 | **Binary Tree Level Order Traversal** | State the cue, invariant, edge case, and proof before implementation. |
| 2 | **Rotting Oranges** | State the cue, invariant, edge case, and proof before implementation. |
| 3 | **Word Ladder** | State the cue, invariant, edge case, and proof before implementation. |
| 4 | **01 Matrix** | State the cue, invariant, edge case, and proof before implementation. |
| 5 | **Open the Lock** | State the cue, invariant, edge case, and proof before implementation. |

## Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

## Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS

REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.